

2021-Ph-H1-Q9

Section: Our Dynamic Universe

Topic: Gravitation

What is the gravitational field strength at a height h above Earth?

Solution:

Use:

$$g = \frac{GM}{(R + h)^2}$$

Gravitational field strength **decreases with distance squared**.

Answer: A

Guidance for Students:

Gravitational field strength weakens as you move away from the planet. Use the inverse square law.

Revision Tip:

Know the formula for gravitational field strength:

$$g = \frac{GM}{r^2}$$

and don't forget to add the Earth's radius to altitude.